

Lab 8

[valid 2024-2025]

JDBC - World Cities

Write an application that allows to **connect** to a relational database by using JDBC, **submit** SQL statements and display the results.

The main specifications of the application are:

Compulsory (1p)

- Create a relational database using any RDBMS (Oracle, Postgres, MySQL, Java DB, etc.).
 - Write an SQL script that will create the following tables:
 - *countries*: id, name, code, continent
 - *continents*: id, name
 - Update *pom.xml*, in order to add the *database driver* to the project libraries.
 - Create a *singleton* class in order to manage a connection to the database.
 - Create *DAO* classes that offer methods for creating countries and continents, and finding them by their ids and names;
 - Implement a simple test using your classes.
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Homework (2p)

- Use a *connection pool* in order to manage database connections, such as [C3PO](#), [HikariCP](#) or [Apache Commons DBCP](#).
 - Create the necessary table in order to store *cities* in your database. A city may contain: id, country, name, capital(true/false), latitude, longitude
 - Create an object-oriented model of the data managed by the application.
 - Create a *tool* to import data from a real dataset: [World capitals gps](#) or other.
 - Display the [distances](#) between various cities in the world.
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Bonus (2p)

- Two cities are *connected* (*sister*, *twins*) if they have a form of legal or social agreement between them, for the purpose of promoting cultural and commercial ties.
Create and insert into your database a large number of fake cities ($\geq 10\,000$) and random sister relationships among them (the sisterhood probability should be low).
- Implement an [algorithm](#) in order to determine all maximal sets of cities with the property that their corresponding subgraphs are 2-connected.
- Create a 2D map (using Swing / JavaFX or generate an image / SVG) representing the cities at their corresponding [locations](#) and their connections. Use different colors for drawing the 2-connected subgraphs.

Resources

- [Slides](#)
- [JDBC](#)
- [Oracle Database JDBC Developer's Guide and Reference](#)
- [JDBC Tutorial - The ULTIMATE Guide](#)

Objectives

- Understand terms and concepts related to relational databases: DBMS, SQL, table, query, stored procedure, cursor, etc.
- Connect to a relational database by using a JDBC driver
- Submit queries and get results from the database
- Specify JDBC driver information externally
- Perform CRUD operations by using the JDBC API